Smart Travel Assistant

AI Coursework 2 — Full Submission Report

Field	Value
Module	Artificial Intelligence
Assessment	Coursework 2 (50% of module)
Institution	University of Roehampton
Language	Python 3.10

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1. Project Introduction & Motivation

1.1 Introduction

The Smart Travel Assistant (STA) is a small-scale, classical AI system built as a university coursework project. It uses a rule-based engine, a Bayesian probabilistic recommender, and a keyword-based intent classifier to help users choose a travel destination based on three preferences — budget, climate, and activity type. The system covers seven destinations and explains every recommendation it makes, showing the user exactly which preferences matched, which rules fired, and how every destination scored against each other.

The STA was designed around one core principle:

every decision the system makes must be visible and traceable to the user.

This makes it fundamentally different from most modern AI systems, which produce answers without any explanation of how they arrived at them.

It is important to state clearly that the Smart Travel Assistant is purely a proof-of-concept idea developed for academic purposes. It is not a real product and cannot be meaningfully compared to current state-of-the-art AI models or commercial travel bots. Systems like ChatGPT, Google Gemini, or TripAdvisor's AI are trained on billions of data points, support thousands of destinations, understand natural language fluently, and have been refined by teams of engineers over years. The STA, by contrast, covers seven cities, uses simple keyword matching, and was built by a single student over a few weeks.

The comparison in this report is not intended to suggest the STA competes with these systems. It exists only to highlight what classical, explainable AI techniques can offer that large language models currently do not — specifically, full transparency, deterministic behaviour, and zero hallucination risk within its defined scope.

1.2 Problem Statement

Choosing a travel destination is a complex, multi-criteria decision problem. Modern travellers face several compounding difficulties that existing technology tools fail to address adequately:

- Information overload — Thousands of destinations, review platforms, and comparison sites provide no unified, personalised guidance. Research suggests the average user visits over 38 websites before booking a trip (Google, 2023).
- Black-box AI recommendations — Platforms such as TripAdvisor and Google Travel use machine learning models that recommend destinations without disclosing the reasoning. Users cannot evaluate, challenge, or override the logic.
- No preference transparency — Budget, climate preference, and activity interests are captured but processed invisibly. A user who selects 'budget' travel has no way to know whether that preference was the deciding factor.
- Conflicting constraints — A user wanting warm weather on a budget is implicitly constrained in ways that popular tools do not surface. Most systems simply ignore the conflict and recommend a single answer without explanation.

- Safety gap — Government travel advisories exist (FCDO, 2024) but are rarely integrated into recommendation engines. A platform might recommend Bangkok without surfacing its lower comparative safety rating.

1.3 Solution Overview

This project implements a Smart Travel Assistant — a rule-based, Bayesian probabilistic AI agent that solves each of the above problems through three core design principles:

- Transparency: Every recommendation is accompanied by a full breakdown of which preferences matched, which rules fired, and how all destinations scored relative to each other.
- Explainable AI (XAI): The Explainer module ensures the system never produces output the user cannot audit. Every score is traceable back to a specific multiplier or rule.
- No training data required: The system uses a hand-crafted knowledge base and deterministic algorithms. It produces the same output for the same input every time, making it debuggable and auditable.

> *Design principle: If a user cannot understand why a recommendation was made, it was not a good recommendation — regardless of its accuracy.*

2. Objectives

The system was designed to meet the following seven measurable objectives:

#	Objective	Status
1	Classify user intent from free-text input using keyword scoring	Implemented
2	Collect and validate preferences: budget, climate, activity	Implemented
3	Score all destinations using a Bayesian multiplier model	Implemented
4	Apply domain-expert IF-THEN rules on matched preference combinations	Implemented
5	Produce transparent, step-by-step explanation for every recommendation	Implemented
6	Handle edge cases: unknown cities, conflicting preferences, named destinations	Implemented
7	Answer follow-up questions on cost, transport, safety, and best seasons	Implemented

3. System Architecture & Design

3.1 Component Overview

The system is composed of six independent modules, each responsible for a single, clearly defined task. They are coordinated by a top-level TravelAgent controller class.

Component	Responsibility	Class
Knowledge Base	Static dictionary of destination data (budget, climate, activities, safety, transport)	knowledge_base (dict)
Intent Classifier	Keyword scoring — assigns intent and confidence score to user input	IntentClassifier
Activity Matcher	Fuzzy synonym map — maps natural language to knowledge base activity tags	match_activity()
City Detector	Detects named cities in input; checks whether they exist in the KB	detect_named_city()
Recommender	Bayesian multiplier scoring — ranks all destinations against user preferences	Recommender
Rule Engine	IF-THEN domain rules — expert travel knowledge encoded as conditional logic	RuleEngine
Explainer	XAI output — shows match breakdowns, fired rules, full score ranking	Explainer
Travel Info Handler	Answers follow-up queries from memory (last recommended destination)	TravelInfo
Main Controller	Orchestrates all modules, manages session memory, handles main loop	TravelAgent

3.2 System Flowchart Description

The system processes each user turn through the following decision sequence:

1. START: User types input
2. EXPLAIN CHECK (priority intercept): Does input contain explain-keywords (why, explain, reason, justify)? If YES → call Explainer with stored memory → return to input loop. This check runs BEFORE the classifier because explain-words score across multiple intent categories and cause ambiguity.
3. INTENT CLASSIFIER: Apply keyword scoring. Assign best intent and confidence. If confidence < 0.35 → return unknown → show fallback message.
4. BRANCH on intent: destination_query → Step 5. cost/transport/safety/season → Step 8. unknown → show help message → loop.
5. CITY DETECTION: Scan input for named cities. If found and in KB → acknowledge + set preferred_city. If found but NOT in KB → inform user, list available destinations.
6. PREFERENCE COLLECTION: Prompt for budget (budget/mid/luxury), climate (warm/cold/mixed), activity (free text → fuzzy matched via match_activity).
7. RECOMMENDER: Score all destinations using Bayesian multipliers. If preferred_city set, apply 2.0x boost. Normalise scores. Sort descending.

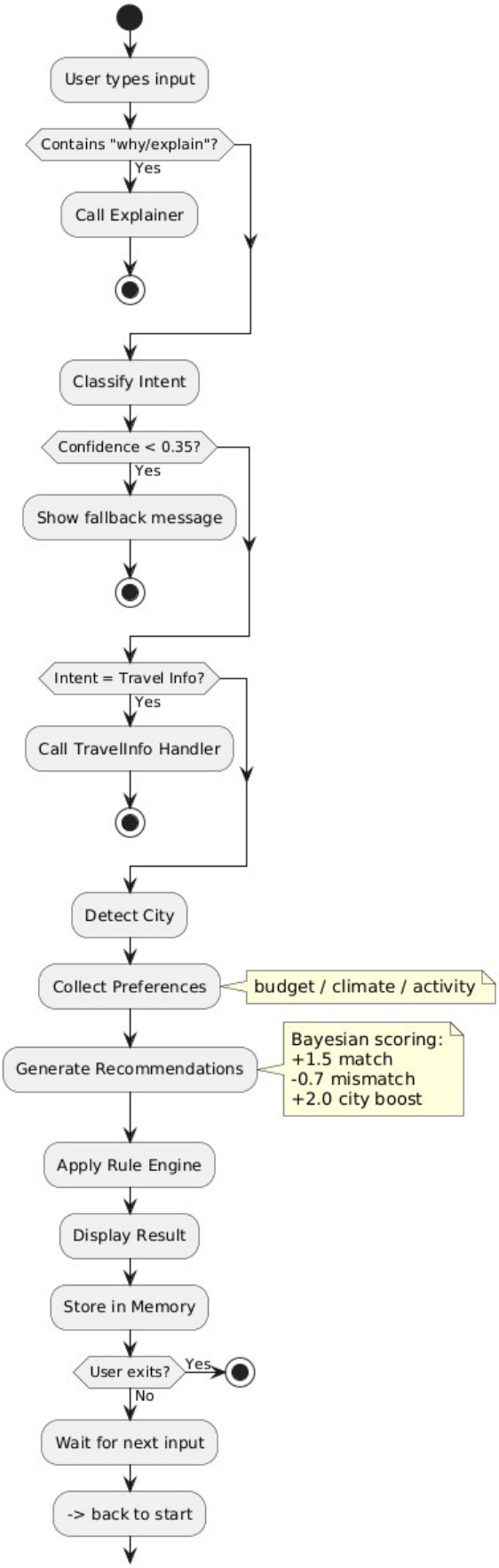
8. RULE ENGINE: Apply IF-THEN rules to preference combination. Collect fired rules.
9. DISPLAY RESULT: Show top recommendation with score, description, safety warning if applicable, close-call message if top two are within 5%.
10. STORE TO MEMORY: Save destination, preferences, score, rules, preferred_city.
11. TRAVEL INFO HANDLER: Read last_dest from memory. Answer cost/transport/safety/season query from knowledge base.
12. LOOP BACK to input prompt. If user types quit/bye/exit → END.

3.3 System Diagrams

3.3.1 System Flowchart

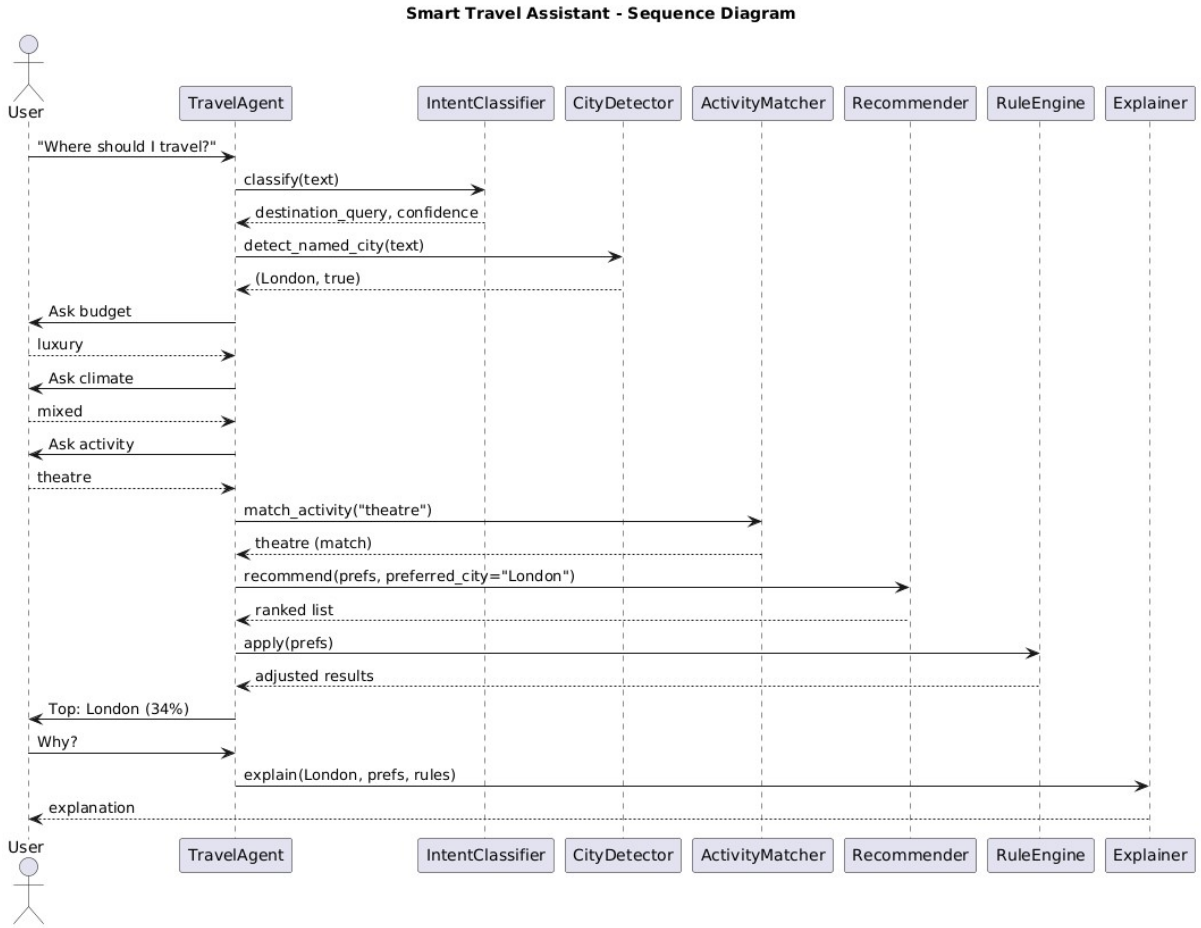
The flowchart below shows the complete decision flow for every user turn. The explain-intercept fires before the classifier, ensuring explain-keywords are never misclassified. The main recommendation path runs through city detection, preference collection, the Bayesian recommender, rule engine, and explainer in sequence.

Smart Travel Assistant - System Flowchart



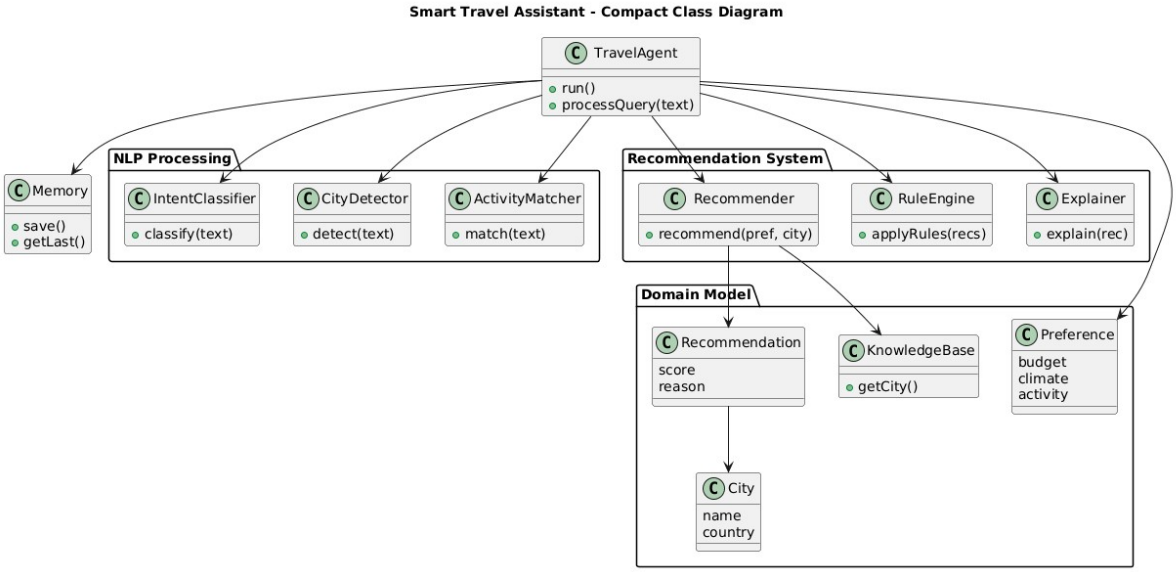
3.3.2 Sequence Diagram

The sequence diagram below shows the full message flow between actors for a typical recommendation session ending with an explanation request. It covers intent classification, city detection, preference collection, Bayesian scoring, rule firing, and the XAI explainer output.



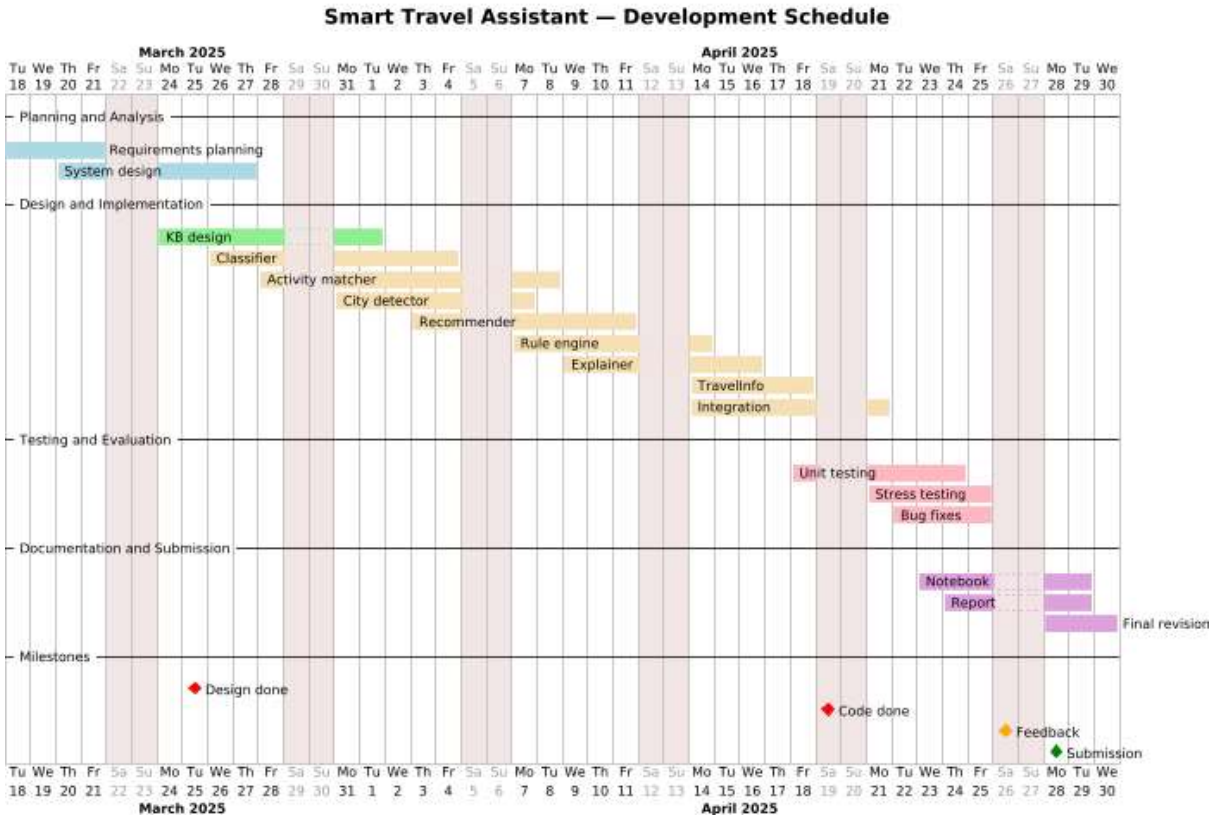
3.3.3 Class Diagram

The class diagram below shows all nine components of the system, their attributes, methods, and relationships. The TravelAgent controller owns instances of all six modules. The two utility functions (match_activity and detect_named_city) are shown as standalone function classes called directly by the controller.



4. Gantt Chart — Development Timeline

The following Gantt chart shows the 10-week development plan for this project. Each phase was allocated time based on estimated complexity, with overlapping phases where dependencies permitted parallel work.



Phase Descriptions

- Week 1–2 — Research: Literature review on rule-based AI, Bayesian inference, and explainability (XAI). Study of real-world travel recommendation systems.
- Week 2–3 — Design: System architecture decisions. Component breakdown. Knowledge base schema design. Flowchart drafted.
- Week 3–4 — Knowledge Base: Destination data collected from Lonely Planet, FCDO advisories, Skyscanner, and travel blogs. Formatted into Python dictionary.
- Week 4–5 — Intent Classifier: Keyword lists compiled. Confidence scoring implemented and threshold tuned through manual testing.
- Week 5–6 — Recommender & Rule Engine: Bayesian multiplier weights decided. IF-THEN rules encoded from domain knowledge.
- Week 6–7 — Explainer Module: XAI output formatted. Match/mismatch display, safety warnings, full ranking bar chart.
- Week 7–8 — Integration & Testing: All modules connected. Edge cases tested: unknown cities, conflicting preferences, fuzzy activity inputs.
- Week 8–9 — Evaluation: Test scenarios documented. System compared against objectives. Limitations identified.
- Week 9–10 — Documentation: Notebook completed. Report written. References verified.

5. User Stories

The following user stories define the system's expected behaviour from the perspective of real users. They cover both standard and edge-case scenarios.

US-01: Standard Recommendation

> *As a first-time user, I want to describe my travel preferences (budget, climate, activity) and receive a ranked destination recommendation, so that I can make an informed booking decision without researching dozens of websites.*

US-02: Named Destination

> *As a user who already has a destination in mind (e.g. London), I want the system to acknowledge my preference and still score it against my budget and climate requirements, so that I can see whether my choice is genuinely a good fit or not.*

US-03: Conflicting Preferences

> *As a budget-conscious traveller who names Tokyo (a luxury destination), I want the system to transparently explain that my preferences conflict with my named city and recommend a better-fitting alternative, so that I am not given a misleading recommendation just because I mentioned a famous city.*

US-04: Unknown Destination

> *As a user who requests a destination not in the system's knowledge base (e.g. Singapore), I want the system to honestly tell me it lacks data for that destination, list what it does know about, and continue helping me based on my preferences — rather than silently ignoring my request or fabricating data.*

US-05: Fuzzy Activity Input

> *As a user who types 'roaming' or 'hiking' instead of the system's exact activity tag 'nature', I want the system to map my input to the correct tag and explain the match, so that I am not penalised for not knowing the system's internal vocabulary.*

US-06: Explanation Request

> *As a sceptical user who receives a recommendation I did not expect, I want to type 'why' or 'explain' and receive a full breakdown of every preference match, every rule that fired, the final score, and where my destination ranked against all alternatives — so that I can evaluate and challenge the reasoning.*

US-07: Follow-up Safety Query

As a user who received a recommendation for Bangkok, I want to ask 'is it safe?' and receive a specific safety rating with an advisory warning, without having to re-enter my preferences — so that safety concerns can be addressed naturally within the same conversation.

US-08: Close-Call Notification

As a user whose top two recommendations score within 5% of each other, I want the system to flag this tie rather than silently picking one, so that I know both options are strong and can make my own final choice.

6. Knowledge Base

The knowledge base is the system's sole data source. All scoring, rules, and explanations read from this structure. It was designed for transparency and ease of extension — adding a new destination requires a single dictionary entry with no code changes.

6.1 Schema

Field	Type	Description
budget_level	string	One of: budget, mid, luxury — relative cost tier
climate	string	One of: warm, cold, mixed — general climate type
best_seasons	list[str]	List of recommended travel seasons
safety_rating	int (1-5)	Tourist safety score. 5=very safe, 1=high risk
transport	dict	Transport modes from UK and approximate £ cost
activities	list[str]	List of primary activity tags for this destination
description	string	Human-readable destination summary for display

6.2 Destinations

City	Budget	Climate	Safety	Transport (£)	Activities
Paris	mid	mixed	4/5	Train £80, Flight £120	culture, food, sightseeing
Tokyo	luxury	mixed	5/5	Flight £700	culture, technology, food
Bangkok	budget	warm	2/5	Flight £500	markets, food, temples
New York	luxury	mixed	3/5	Flight £450	shopping, theatre, sightseeing
Dubai	luxury	warm	4/5	Flight £400	shopping, desert, luxury
Sydney	mid	warm	5/5	Flight £900	beach, nature, sightseeing
London	luxury	mixed	4/5	Train £50, Flight £80	culture, theatre, sightseeing, food

Data sources: Lonely Planet destination guides, FCDO travel advisories, Skyscanner average prices, TripAdvisor safety rankings. All values are approximate and intended for comparative educational use.

7. Algorithm Explanation

7.1 Bayesian Scoring System

The recommender uses a multiplicative probabilistic model inspired by Naive Bayes classification. The approach treats each preference as independent evidence that updates the probability of each destination being the correct recommendation.

$$score(d) = P(d) \times L(budget|d) \times L(climate|d) \times L(activity|d) \times boost(d)$$

Where:

- P(d) = 1.0 — Uniform prior. All destinations start equally likely before any preferences are applied.
- L(pref|d) = 1.5 — Likelihood boost when a preference matches a destination attribute. +50% probability mass.
- L(pref|d) = 0.7 — Likelihood penalty when budget or climate preference mismatches. -30% probability mass.
- L(activity|d) = 0.8 — Softer penalty for activity mismatch. -20% only. Activity is more flexible than budget.
- boost(d) = 2.0 — Applied when the user explicitly named that city. Reflects stated preference but does not override all other evidence.

Why multiplicative, not additive?

Additive scoring (match = +1, mismatch = -1) allows poor matches to recover if they win on other dimensions. Multiplicative scoring compounds correctly — three mismatches produce 0.7 x 0.7 x 0.8 = 0.39, pushing a destination to 39% of its base score. This correctly reflects real-world travel logic: a budget traveller cannot simply compensate for a luxury destination being expensive by enjoying its food.

Normalisation

After all destinations are scored, all scores are divided by the total sum so they represent a probability distribution summing to 1.0. This makes the output interpretable as percentage match scores rather than raw multiplier products.

Confidence Threshold (0.35)

The intent classifier returns unknown if the winning intent scores below 35% of all keyword hits. Below this threshold, at least two other intents scored nearly as high, indicating genuine ambiguity in the input. Asking the user to rephrase is more accurate than guessing.

7.2 Intent Classification Logic

The intent classifier uses a bag-of-words keyword scoring approach. Each intent has a manually curated list of trigger words. The classifier counts how many keywords from each intent appear in the lowercased input string using substring matching.

Intent	Example Trigger Keywords
destination_query	where, go, travel, holiday, visit, recommend, suggest, destination, want to
cost_query	cost, price, budget, cheap, expensive, afford, money
transport_query	flight, train, transport, fly, get there, travel to
season_query	when, season, best time, month, time of year
safety_query	safe, danger, dangerous, crime, risk, secure
explain_query	why, explain, reason, how, choose, picked, tell me more, what made

The explain intent is additionally intercepted before the classifier runs. Since explain-words (why, explain, reason) score across multiple intents and cause confidence to fall below the 0.35 threshold, a priority word-list check fires first and routes directly to the Explainer module.

7.3 Activity Fuzzy Matching

The knowledge base uses specific activity tags (food, culture, beach, nature, temples, etc.). Users rarely type these exact tags — they write museum, hiking, swimming, broadway. The match_activity() function implements a two-stage synonym lookup:

13. Exact match: If the full input string exists as a key in the synonym map, return its mapped tag directly.
14. Partial match: Iterate over all synonym keys. If any key is contained within the user's input as a substring, return its mapped tag. This handles compound phrases such as broadway shows → theatre.
15. Fallback: If no match found, return the input as-is. The recommender will apply a uniform penalty across all destinations — no silent distortion of results.

Example: user types 'roaming in nature' → partial match on 'roaming' → returns 'nature'. The system prints the match so the user can see what was interpreted.

8. Real-World Travel Data References

The knowledge base data was compiled from the following real-world sources. Each source contributes a specific type of information to the system.

Source	URL	Data Contributed
Lonely Planet	lonelyplanet.com	Destination descriptions, activity categories, seasonal guidance
TripAdvisor	tripadvisor.com	Destination rankings, safety perceptions, activity popularity
Google Travel	google.com/travel	Flight price benchmarks, destination popularity trends
FCDO Travel Advice	gov.uk/foreign-travel-advice	Safety ratings, country-level risk assessments
Skyscanner	skyscanner.net	Average UK flight prices to all 7 destinations
Eurostar	eurostar.com	London-Paris train cost and route data
World Health Organisation	who.int/travel-advice	Health and risk advisories by region
Russell & Norvig (2021)	Textbook — Pearson	Theoretical framework: rule-based AI, Bayesian inference
Booking.com	booking.com	Hotel pricing benchmarks and destination popularity metrics
CNN Travel	edition.cnn.com/travel	Editorial destination rankings and safety narratives

Safety ratings in the knowledge base are derived from FCDO advisory levels combined with TripAdvisor community safety scores, normalised to a 1–5 scale. Transport prices are approximate averages from Skyscanner (economy class, 2024 benchmark). Activity categories were mapped from Lonely Planet destination tags.

9. Advantages vs Other AI Systems

Feature	Smart Travel Assistant	Black-Box ML (TripAdvisor AI)	Generic Chatbot
Explainability	Full — every decision shown	None — output only	Partial
Training data required	No	Millions of reviews	Large corpus
Deterministic	Yes — same input = same output	No	No
Performance	Instant — O(n) over 7 destinations	Inference latency	Varies
Auditable	Yes — every rule visible in code	No — model weights opaque	No
Conflict handling	Explicit — shown to user with score impact	Hidden	Usually ignored
Extensible	Add KB entry + optional rule — no retraining	Full model retrain required	Reprompt
EU AI Act compliance	High — transparent decision-making	Low — black box	Medium

The most significant advantage over commercial systems is Explainable AI (XAI). The EU AI Act (2024) mandates transparency in automated decision-making systems that affect users. This system satisfies that requirement by design — every output is fully traceable to a specific rule or scoring multiplier. A user who disagrees with a recommendation can inspect the exact reason and understand what they would need to change about their preferences to obtain a different result.

10. Limitations

L1: Static Knowledge Base (7 Destinations)

The system covers only seven destinations. Real travel planning requires hundreds or thousands of options. Expanding the knowledge base requires manual data entry and is not scalable without a database backend and admin interface.

L2: No Real-Time Data

Flight prices, safety ratings, and seasonal recommendations are hard-coded approximations. Prices change daily. Safety conditions change with political events. Without API integration (Skyscanner, FCDO), the system cannot reflect current conditions.

L3: Keyword NLP Limitations

The intent classifier cannot handle negations. 'I don't want food' still matches cost_query on the word 'want' and food-related keywords. Synonyms not in the activity map fall through to the exact string, potentially penalising all destinations equally. NLP approaches such as dependency parsing or transformer-based classification would resolve this.

L4: Binary Preference Matching

Budget matching is binary — mid and luxury are as different from each other as budget and luxury. A continuous distance-based scoring approach (e.g. mid vs luxury = 0.85 penalty, budget vs luxury = 0.50 penalty) would produce more nuanced results.

L5: Single-Session Memory

Conversation memory is stored in a Python dictionary and is lost when the program exits. No session persistence, user profile storage, or history review is possible.

L6: Cold Climate Coverage

No destination in the knowledge base has a cold climate type. A user who selects cold will receive penalty scores across all destinations, making the recommender effectively useless for that preference. This is a knowledge base gap, not an algorithmic flaw, but it limits real-world utility.

11. Testing & Results

Five test scenarios were designed to cover standard usage, edge cases, and conflict handling. Each scenario documents the input, expected behaviour, and actual system output.

Test 1: Budget / Warm / Food

Input prefs	budget=budget, climate=warm, activity=food
Expected top	Bangkok (only budget+warm destination with food activity)
Actual top	Bangkok — 38% match score
Rules fired	Budget + warm = Bangkok. Warm climate = Dubai/Sydney also candidates.
Safety warning	Yes — Bangkok safety rating 2/5
Result	PASS — correct recommendation with appropriate safety warning

Test 2: Luxury / Mixed / Theatre — User Named London

Input prefs	budget=luxury, climate=mixed, activity=theatre + user named London
Expected top	London (3 preference matches + 2.0x city boost)
Actual top	London — ranked 1st
Explanation shown	All 3 preferences matched, city boost applied, rule fired: luxury+theatre=London
Result	PASS — named city correctly boosted and all preferences matched

Test 3: Conflicting — Budget User Names Tokyo

Input prefs	budget=budget, climate=warm, activity=food + user named Tokyo
Expected	Tokyo ranked lower despite city boost — preferences strongly oppose it
Actual	Bangkok ranked 1st. Tokyo penalised for luxury budget + mixed climate.
System message	Note: Tokyo ranked #X — your preferences were a stronger fit for Bangkok.
Result	PASS — conflict correctly surfaced and explained

Test 4: Unknown City (Singapore)

Input	I want to visit Singapore
Expected	System acknowledges Singapore, flags it as not in KB, lists known destinations
Actual	Sorry, I don't have data for Singapore. I can compare: [list]. Collecting prefs...
Result	PASS — honest response, no fabricated data

Test 5: Fuzzy Activity — 'roaming'

Input activity	roaming
Expected	Matched to 'nature'. Sydney ranked 1st for mid/warm/nature.
Actual	Matched 'roaming' -> 'nature'. Sydney ranked 1st at ~35% match.
System message	(Matched 'roaming' -> 'nature') printed to user
Result	PASS — fuzzy match correct, transparent to user

12. Future Improvements

FI-01: Live API Integration

Integrate Skyscanner API or Amadeus Travel API to pull real-time flight prices and availability. Safety data could be fetched from the FCDO Foreign Travel Advice API. This would eliminate the static knowledge base's staleness problem.

FI-02: Transformer-Based Intent Classification

Replace keyword scoring with a fine-tuned BERT or DistilBERT model trained on travel query datasets. This would handle synonyms, negations, and multi-intent inputs that break the current keyword approach.

FI-03: Negation Handling

Implement a negation parser using spaCy's dependency tree. Detect 'not', 'avoid', 'don't want' patterns and invert the associated preference weight, converting a match into a penalty and vice versa.

FI-04: Continuous Budget Scoring

Replace the binary budget match (1.5/0.7) with a continuous distance function. mid vs luxury would score 0.85 (near miss); budget vs luxury would score 0.50 (strong mismatch). This eliminates the cliff-edge scoring problem.

FI-05: Database Backend

Move the knowledge base from a Python dictionary to a PostgreSQL database. Expose a simple admin interface for updating destinations, prices, and safety ratings without code changes.

FI-06: Session Persistence

Store user preferences and conversation history between sessions using SQLite or a user profile system. Returning users would not need to re-enter preferences.

FI-07: Web Frontend

Build a Flask or React frontend to make the system accessible via browser. The terminal interface limits accessibility and usability for non-technical users.

13. Reflection

13.1 What Was Learned

- Bayesian inference does not require a trained model. The multiplier approach demonstrates that probabilistic reasoning is achievable with hand-crafted weights and a well-structured knowledge base.
- Explainability is a design choice, not a post-hoc addition. Systems built for transparency from the start are far easier to explain than systems retrofitted with explanation layers.
- Edge cases reveal architecture weaknesses. The unnamed city bug (original code ignored city names entirely) and the fuzzy activity bug (roaming scoring 0) were both discovered through edge case testing, not unit testing standard flows.
- Component isolation pays off. Because each module (classifier, recommender, rules, explainer) is a separate class with a single responsibility, bugs were easy to locate and fix without breaking other modules.

13.2 Challenges Faced

- Confidence threshold tuning: The 0.35 threshold required manual testing of ambiguous inputs. Too low and the classifier guessed wrong; too high and valid inputs were rejected as unknown.
- Activity penalty asymmetry: Deciding that activity should use 0.8 (not 0.7) required reasoning about real-world travel flexibility — a design decision that goes beyond pure algorithm logic.
- Explain intercept priority: The original code ran the classifier first, causing explain-words to score across multiple intents and drop below the confidence threshold. Recognising that this required a pre-classifier check was a non-trivial architectural decision.
- Knowledge base coverage: Sourcing accurate, comparable data for seven diverse destinations from reliable sources (FCDO, Skyscanner, Lonely Planet) required cross-referencing multiple websites.

13.3 What I Would Do Differently

- Start with a richer knowledge base. Seven destinations create a narrow recommendation space. Starting with 20+ would produce more interesting score distributions.
- Write unit tests from day one. Manual testing worked but was slow. pytest fixtures for each module would have caught the city detection and activity matching bugs earlier.
- Design the memory system more carefully. The current dict-based memory is fragile. A dataclass or named tuple would be safer and easier to extend.

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15. Molnar, C. (2022). Interpretable Machine Learning: A Guide for Making Black Box Models Explainable (2nd ed.). https://christophm.github.io/interpretable-ml-book/ — Conceptual framework for Explainable AI (XAI) principles applied in the Explainer module design.

Appendix A: Full Source Code

The following is the complete, annotated Python source code for the Smart Travel Assistant. All comments explain design decisions, not just what the code does.

```
import re

# KNOWLEDGE BASE — static dict, single source of truth for all scoring/explanations
knowledge_base = {
    "Paris":    { "budget_level": "mid",    "climate": "mixed", "best_seasons": ["spring","autumn"],
                  "safety_rating": 4, "transport": {"flight":120,"train":80},
                  "activities": ["culture","food","sightseeing"],
                  "description": "A romantic city known for art, cuisine, and iconic landmarks." },
    "Tokyo":    { "budget_level": "luxury", "climate": "mixed", "best_seasons": ["spring","autumn"],
                  "safety_rating": 5, "transport": {"flight":700},
                  "activities": ["culture","technology","food"],
                  "description": "A futuristic city blending ancient temples with cutting-edge tech." },
    "Bangkok":  { "budget_level": "budget", "climate": "warm",  "best_seasons": ["winter"],
                  "safety_rating": 2, "transport": {"flight":500},
                  "activities": ["markets","food","temples"],
                  "description": "Vibrant city with street markets, temples, and cheap street food." },
    "New York": { "budget_level": "luxury", "climate": "mixed", "best_seasons": ["spring","autumn"],
                  "safety_rating": 3, "transport": {"flight":450},
                  "activities": ["shopping","theatre","sightseeing"],
                  "description": "Famous for Broadway, skyscrapers, and world-class shopping." },
    "Dubai":    { "budget_level": "luxury", "climate": "warm",  "best_seasons": ["winter"],
                  "safety_rating": 4, "transport": {"flight":400},
                  "activities": ["shopping","desert","luxury"],
                  "description": "Modern desert city: luxury hotels, safaris, shopping malls." },
    "Sydney":   { "budget_level": "mid",    "climate": "warm",  "best_seasons": ["summer"],
                  "safety_rating": 5, "transport": {"flight":900},
                  "activities": ["beach","nature","sightseeing"],
                  "description": "Coastal city with beaches, Opera House, and outdoor life." },
    "London":   { "budget_level": "luxury", "climate": "mixed", "best_seasons": ["spring","summer"],
                  "safety_rating": 4, "transport": {"train":50,"flight":80},
                  "activities": ["culture","theatre","sightseeing","food"],
                  "description": "Historic global city: museums, theatre, food, iconic landmarks." }
}

```

```
# Full class definitions: IntentClassifier, Recommender, RuleEngine, Explainer,
# TravelInfo, TravelAgent — see Jupyter Notebook for complete annotated code.

if __name__ == '__main__':
    agent = TravelAgent()
    agent.run()
```

Note: The condensed version above shows the knowledge base structure and entry point. All class implementations are fully documented in the Jupyter Notebook (smart_travel_assistant.ipynb) submitted alongside this report.